Pratik Ganesh Meshram

Computer Science Student

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EDUCATION:

VISHWAKARMA INSTITUTE OF TECHNOLOGY PUNE

Bachelor Of Technology in CSE

2022 - 2026

- CGPA - 8.33
- SGPA - 9.8

NOWROSJEE WADIA COLLEGE PUNE

Higher Secondary (Class XII)

2021 - 2022

DR. MRS. ERIN N. NAGARVALA DAY SCHOOL PUNE

Secondary School (Class X)

2019 - 2020

TECHNICAL SKILLS:

- **Programming Languages :**
  C / C++, Python
- **Data Structures & Algorithms :**
  Arrays, Linked Lists, Trees, Sorting & Searching Algorithms, Recursion
- **Machine Learning:**
  Linear Regression, Logistic Regression, Decision Trees, Random Forest, Support Vector Machine (SVM), Naive Bayes
- **Deep Learning:**
  Artificial Neural Network, Convolutional Neural Network
- **Data Analysis & Visualization:**
  Pandas, Numpy, Matplotlib
- **Database & Tools:**
  SQL, Jupyter Notebook, Git & GitHub

SUMMARY:

Motivated Computer Science graduate with a strong foundation in Artificial Intelligence, Machine Learning, and Data Science. Experienced in developing predictive models, data preprocessing, and exploratory data analysis using Python, Scikit-learn, Pandas, and NumPy. Passionate about solving real-world problems through AI and seeking an entry-level AI/ML role to contribute technical expertise while continuously learning and growing in the field.

ACADEMIC PROJECTS:

PLANKTON AI – AUTONOMOUS PLANKTON DETECTION SYSTEM — 2025

SIH (Smart India Hackthon) 1st Winning Project

- Designed and implemented a Computer Vision pipeline for plankton detection and species classification from microscopic images.
- Built an end-to-end image processing pipeline including image preprocessing, augmentation, and feature extraction techniques, using YOLO v8n and Mobile Net v3.
- Evaluated classification models using Accuracy, Precision, Recall, and F1-Score, enabling reliable automated analysis.

AERO FLAME FIRE SURVEILLANCE DRONE — 2024

ESP32 + OpenCV + BetaFlight | Drone-based IoT Project

- Developed a real-time fire/flame detection system using an Ultralytics YOLO model in Python.
- Processed each video frame with OpenCV (resizing to 640×480) and performed object detection; then drew bounding boxes with confidence scores around detected fires.
- Integrated the YOLO pipeline into a Flask web app for live video upload and streaming inference, enabling automated fire alerts in the web interface.

ANIMAL TRACKING & HEALTH MONITORING BELT — 2023

Provisonal Patent, Department Winner Project, Hackthon Project

- Designed time-series sensor data pipeline with Adafruit IO & Firestore to support ML-based predictive health modeling for early illness detection
- Implemented on-device edge geofencing on ESP32 with automated Twilio SMS alerts, eliminating cloud dependency for real-time breach detection
- Developed health-production correlation analytics module mapping milk yield trends against live vitals for data-driven disease detection insights.

ACHIEVEMENTS:

- **Awards/Activities:** SIH Winner (Plankton AI – Autonomous Plankton Detection System), Presented Animal Tracking And Helth Monitoring Belt DXC company).
- **Certifications:** PW Skills (Decode DSA), Udemy Ethical Hacking).